

$n\text{-SrTiO}_3$ absorber

$$0 \leq x \leq L$$

Poisson equation

electron and hole drift-diffusion

uniform photogeneration G

SRH recombination R_{SRH}

— electrons n

— holes p

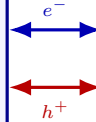
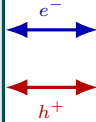
$$E_{\text{CB,s,O}}^0 - E_{\text{CB,s,H}}^0 = +0.150 \text{ eV}$$

HER-side catalyst

(100) facet,

$$i = H, E_{\text{cat,HER}}$$

Catalysis: HER/HOR



OER-side catalyst

(110) facet,

$$i = O, E_{\text{cat,OER}}$$

Catalysis: OER/ORR

$$i = H, O: \quad j_{n,i} = qk_n(n_{s,i} - n_{\text{eq},i}) \quad j_{p,i} = qk_p(p_{s,i} - p_{\text{eq},i})$$

Adaptive contact (A)

$E_{\text{CB,s,i}} = E_{\text{CB,s,i}}^0, \quad \Phi_{B,i} = E_{\text{CB,s,i}}^0 - E_{\text{cat,i}}$
Catalyst polarization changes the effective transfer barrier while the STO surface band remains pinned.

Buried Schottky contact (B)

$\Phi_{B,i} = \Phi_{B,i}^0, \quad E_{\text{CB,s,i}} = E_{\text{cat,i}} + \Phi_{B,i}^0$
Catalyst polarization shifts the adjacent STO bands while the Schottky barrier remains fixed.